CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2020 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Explain why Java is considered the best language for Internet Applications.

Reasons: Platform independence via bytecode/JVM, built-in security (sandbox security manager), robust memory management, native multithreading, and extensive networking APIs (`java.net`).

Q2 (3M) Discuss various primitive data types supported by Java.

Primitive Types: 8 primitive types: `byte` , `short` , `int` , `long` , `float` , `double` , `char` , `boolean` with explicit fixed sizes across all platforms.

Q3 (3M) What is the purpose of the 'static' keyword in Java?

Explanation: Marks class-level members (variables, methods, blocks) shared by all instances, callable without creating an object of the class (e.g., `ClassName.staticMethod()`).

Q4 (3M) What are Wrapper Classes in Java?

Explanation: Classes that wrap primitive data types into object representations (`Integer` , `Double` , `Boolean`) enabling autoboxing and unboxing for collections.

Q5 (3M) Define Interface in Java.

Definition: A reference type containing pure abstract method declarations and static constants, defining a functional contract for implementing classes.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Applet Life Cycle and write an Applet to display a greeting message.

Applet Lifecycle: Complete coverage of `init()` , `start()` , `paint()` , `stop()` , and `destroy()` .

Q7 (7.5M) Explain Thread Synchronization and the 'synchronized' keyword in Java.

Synchronization: Prevents thread interference and race conditions when multiple threads access shared resources simultaneously using monitor locks.

Q8 (7.5M) Differentiate between Throw and Throws in Java Exception Handling.

Throw: Explicitly triggers an exception instance inside a method body.
Throws: Declares exceptions thrown by a method in its header signature.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is JDBC? Explain JDBC Architecture and execute a database SELECT query.

JDBC Architecture: Application → JDBC API → DriverManager → JDBC Driver → Database. Complete sample SELECT query program.

Q10 (15M) **Write HTML code for a complete Web Form with Table formatting and Validation.**

HTML Implementation: Form markup with input types, select dropdowns, radio buttons, and table alignments.

Q11 (15M) **Discuss JSP Architecture, Scripting Elements (<%! %>, <% %>, <%= %>), and Directives.**

JSP Scripting Elements:

- **Declaration (`<%! %>`):** Declares class variables and methods.
- **Scriptlet (`<% %>`):** Java code placed inside the `_jspService()` method.
- **Expression (`<%= %>`):** Directly prints evaluated values to the output stream.

Q12 (15M) **Explain Servlets, HttpServletRequest, and HttpServletResponse with complete GET/POST demonstration.**

Servlet Processing: Complete explanation of HTTP request parsing, header inspection, and HTML output streaming.

Q13 (15M) **What are JavaBeans? Explain Bean properties and deployment in JSP using jsp:useBean.**

JavaBeans: Reusable software components adhering to conventions: zero-arg constructor, private fields, and public getters/setters.
